DATA 205 PROJECT PLAN
MARYLAND STATE AID ANALYSIS (FY2009–2024)

Distribution Patterns and Economic Shifts

Ike Depenyou

Data205

DATA OVERVIEW

Source: Maryland Open Data Portal

Time span: 2009–2024

Includes: Grants & Loans ≥ $50,000

Key variables:

- Fiscal Year
- Grantor (agency)
- Amount
- Zip Code
- Category (Grant vs Loan)

What to say:
"The dataset contains over 15 years of state aid records, focusing on

DATA LIMITATIONS

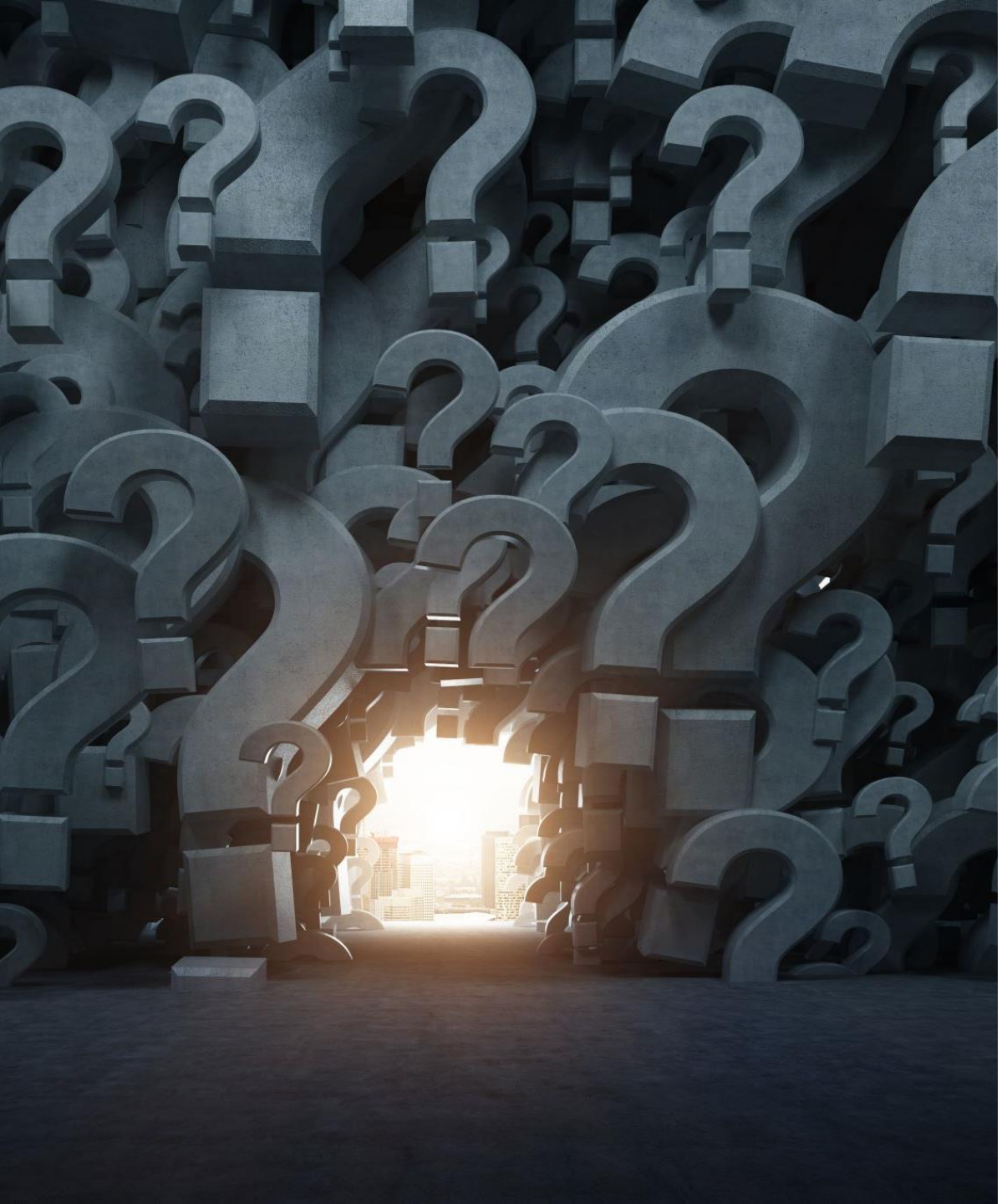
- **Content:**
- Left-truncated (only ≥ $50,000)
- Bias toward large organizations
- Inconsistent naming (e.g., agency variations)

"One limitation is that smaller awards are excluded, which may bias results. Additionally, inconsistencies in naming require extensive data cleaning."

RESEARCH GOAL

- **Primary Goal:**
  Analyze distribution patterns and shifts in economic priorities
- **Focus Periods:**
- Pre-pandemic (2017–2019)
- Pandemic (2020–2022)
- Post-pandemic (2023–2024)

"The central objective is to understand how funding priorities evolved, especially during the COVID-19 crisis."



KEY RESEARCH QUESTIONS

- Annual grant volume trends
- Top funding agencies
- Grant vs loan differences
- COVID impact on recipients
- Geographic distribution
- Forecast for 2025

"These questions guide both descriptive and predictive analysis, allowing us to examine trends, disparities, and future projections."

METHODS AND MODELS

- T-test → Grant vs Loan comparison
- Regression → Predict award amount
- K-means → Geographic clustering
- ARIMA → Forecasting
- PCA → Funding concentration

"I apply a combination of statistical and machine learning methods to answer different types of questions, from hypothesis testing to forecasting."

TOOLS USED & DATA PRODUCT

- R → Data cleaning, statistical analysis, and modeling
- Tableau → Interactive visualization
- Interactive Tableau dashboard
- Filters:
  - Year
  - Agency
- Includes narrative "data story"

"The final output is an interactive dashboard that allows users to explore how funding varies by year and agency. It also includes a narrative component highlighting shifts in funding priorities during the COVID-19 period."

PROJECT TIMELINE

- **Content:**
- Data cleaning (R)
- Exploratory data analysis
- Statistical modeling (R)
- Dashboard development (Tableau)
- Final presentation

"The project follows a structured workflow beginning with data preparation in R, followed by analysis and modeling, and concluding with visualization and presentation."

THANK YOU GUYS ANY QUESTIONS ?